11.1 Atoms, Nuclei & Radiation

Question Paper

Course	CIE A Level Physics
Section	11. Particle Physics
Topic	11.1 Atoms, Nuclei & Radiation
Difficulty	Hard

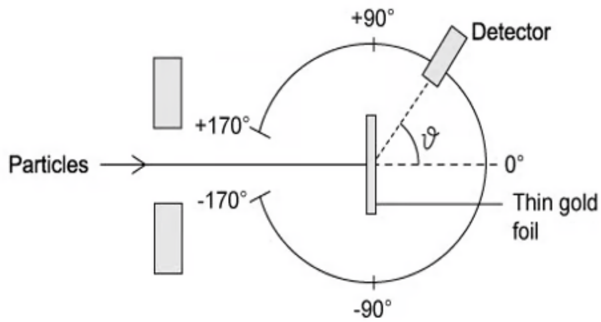
Time allowed: 10

Score: /4

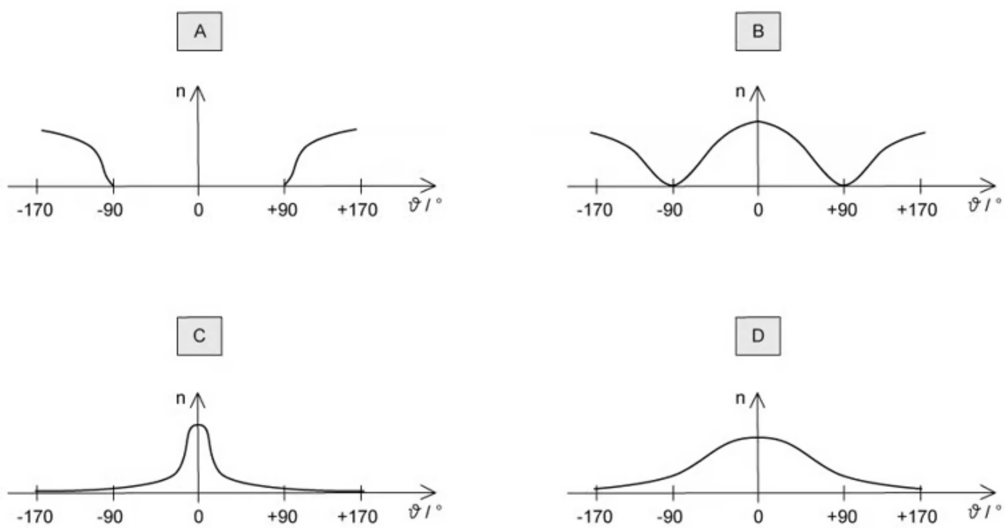
Percentage: /100

Question 1

In an α-particle scattering experiment, a student set up the apparatus below to determine the number *n* of α-particles incident per unit time on a detector held at various angles θ.



Which of the following graph best represents the variation of *n* with θ?



[1 mark]

Question 2

A stationary decay ${}^{210}_{84}Po$ of by α-emission has a total kinetic energy of E_k

$${}^{210}_{84}Po \rightarrow {}^{206}_{82}Pb + \alpha$$

What is the kinetic energy of the daughter nucleus ${}^{206}_{82}Pb$?

A. 0

B. between 0 and $\frac{E_k}{2}$

C. between $\frac{E_k}{2}$ and E_k

D. E_k

[1 mark]

Question 3

Which of the following equations correctly shows an α-particle causing a nuclear reaction ?

A. ${}^{14}_{7}N + {}^{4}_{2}He \rightarrow {}^{17}_{8}O + {}^{1}_{1}n$

B. ${}^{17}_{8}O + {}^{4}_{2}He \rightarrow {}^{20}_{9}F + {}^{1}_{1}P$

C. ${}^{17}_{8}O + {}^{0}_{-1}e \rightarrow {}^{13}_{5}B + {}^{4}_{2}He$

D. ${}^{14}_{7}N + {}^{1}_{1}p \rightarrow {}^{11}_{6}C + {}^{4}_{2}He$

[1 mark]

Question 4

Thorium $^{232}_{90}Th$ decays through a series of transformations. The particles emitted in successive transformations are

$\alpha \ \beta \ \beta \ \gamma \ \alpha$

The resulting nuclide may be represented by

A. $^{230}_{82}Pb$

B. $^{227}_{85}At$

C. $^{225}_{87}Fr$

D. $^{224}_{88}Ra$

[1 mark]

Model Answers: Hard

1

The correct answer is **C** because:

- The gold foil experiment directed parallel beams of α-particles at a piece of gold foil only 10^{-6} m thick
- The observations were as follows:
  - Most of the α-particles went straight through the foil - so, the largest value of *n* is at small angles
  - Some of the α-particles were deflected slightly - here, *n* drops quickly with increasing angle of deflection θ
  - Some α-particles, about 1 in 20 000, were deflected by more than 90°, appearing to bounce back off the foil - so, the value of *n* is very small at large angles
- Therefore, the graph that matches this description is **C**

A & B are incorrect because these graphs show the majority of α-particles were deflected through large angles which is the opposite of what was observed

D is incorrect because this graph shows many particles were deflected through a variety of angles, however, the observations show that the majority of particles passed straight through the foil with little to no deflection

2

The correct answer is **B** because:

- By the Principle of Conservation of Momentum :
- Total momentum before α-emission = Total momentum after α-emission
  - Total momentum before α-emission = 0
  - Total momentum after α-emission $= MV - mv$
- Where M and V are the mass and velocity of ${}^{206}_{82}Pb$ and m and v are the mass and velocity of the α-particle
  - $0 = MV - mv$
  - $MV = mv$
  - $M^2V^2 = m^2v^2$
  - $M\left(\frac{1}{2}M\ V^2\right) = m\left(\frac{1}{2}mv^2\right)$
- Therefore, $\frac{KE\ of\ Pb-206}{KE\ of\ \alpha} = \frac{mass\ of\ \alpha}{mass\ of\ Pb-206}$
  - $\frac{KE(Pb)}{KE(\alpha)} = \frac{4}{206} = 0.0194$
- From Conservation of Energy : $KE(Pb) + KE(\alpha) = E_k$
  - $KE(\alpha) = \frac{KE(Pb)}{0.0194}$
  - $KE(Pb) + \frac{KE(Pb)}{0.0194} = E_k$
  - $KE(Pb)\left[1 + \frac{1}{0.0194}\right] = E_k$
  - So, $KE(Pb) = \frac{E_k}{\left(1 + \frac{1}{0.0194}\right)} = 0.0190\ E_k$
- Therefore, the kinetic energy of ${}^{206}_{82}Pb$ is between 0 and $\frac{E_k}{2}$

3

The correct answer is **B** because:

- An α-particle is a helium nucleus, which is represented by ^4_2He
- A neutron is represented by ^1_0n
- Since the α-particle is supposed to cause the reaction, it should appear on the left of the equation as a reactant
  - Hence, only option **A** or **B** can be correct
- In option **A**, the symbol of the neutron is wrongly represented as ^1_1n
  - Therefore, since option **A** is not possible, this leaves only option **B**
- We can confirm this by comparing the nucleon numbers on each side
  - LHS: 17 + 4 = 21
  - RHS: 20 + 1 = 21
- Nucleon number is conserved, so option **B** is correct

A is incorrect because the symbol of the neutron is wrongly represented as ^1_1n

C & D are incorrect because the α-particle causes the reaction, so, it should appear on the left of the equation as a reactant

4

The correct answer is **D** because:

- During α-decay, a helium nucleus ^4_2He is released from each starting atom
  - The atomic number decreases by 4 and the proton number by 2
- During β radiation, an electron (or positron) is released
  - There is a gain of 1 proton because a neutron splits to form a proton and electron
  - There is no change in atomic number
- During γ radiation, γ rays are emitted but there is no change in atomic or proton number
- We can draw a flow diagram of the processes involved:

$$^{232}_{90}Th \xrightarrow{\alpha} {}^{228}_{88}Ra \xrightarrow{\beta} {}^{228}_{89}Ac \xrightarrow{\beta} {}^{228}_{90}Th \xrightarrow{\gamma} {}^{228}_{90}Th \xrightarrow{\alpha} {}^{224}_{88}Ra$$

- Therefore, the end nuclide is $^{224}_{88}Ra$